

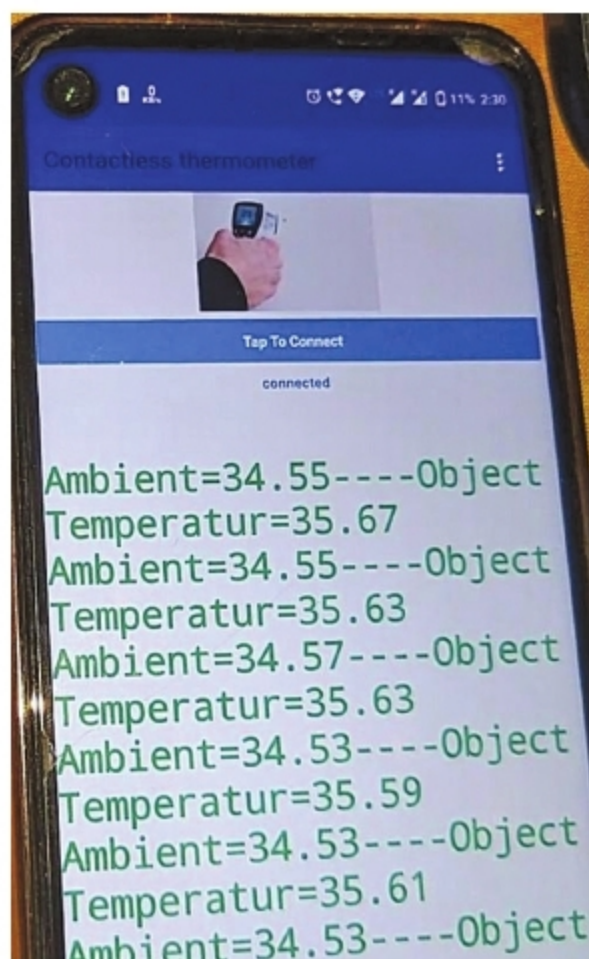


Fig. 6: Temperature readings on mobile phone

tors such as MIT App Inventor, Kodular, Android studio, Eclipse, and others. For this project, Kodular has been used to create the app.

First, open Kudular on <https://www.kodular.io/> website using a valid email ID. Then, create a new app by clicking the 'Create project' option. Name the project as per your choice. Here, the project name is 'Contactless IR thermometer.' Next, make the app layout and add the following components to the layout:

- Bluetooth client
- File
- Listpicker
- Clock
- Image

There are three sections in the Kodular platform: Assets, Designer, and Blocks. The app layout in the Designer section is shown in Fig. 4. Open the Blocks section and start the



Fig. 7: Connecting OTG with mobile

coding by joining various blocks, as shown in Fig. 5. Next, export the project as IRThermo.apk file and install it on your Android phone.

Testing

After making the connections and installation of the app on your phone, enclose the hardware along with a 9V battery in a suitable plastic box with MLX90614 sensor fitted on the front

EFY Note

The source code
of this project
is available
for free download
at source.efymag.com

panel. But first, you need to upload the Arduino code (IRThermo.ino) to the Arduino board.

Turn on the Bluetooth on your phone, open the app, and click on the 'Tap to Connect' button. Select HC-05 to connect it. After a successful connection, you can see the temperature readings on the app, as shown in Fig. 6.

Keep the IR temperature sensor near your forehead (about 5cm) and check the readings, which should be between 35°C and 37°C. The app can also save the temperature readings in a text file. You can check them in the file manager.

USB based contactless IR thermometer

You can also make the project by omitting the Bluetooth HC-05 module. With the same code in the Arduino board, connect the circuit with the phone using a USB cable and an OTG (on-the-go) connector, as shown in Fig. 7.

You need to install a serial monitor app such as Serial USB Terminal (by Kai Morich) on your phone. After connecting the circuit with the mobile phone, you can see the object and surrounding temperatures on the phone. **EFY**

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